Quantitative EEG and ERP report

CLINIC C LOGO

Patient:	Mock Patient 042
Date of birth:	1 Jan 2016
Age:	10 years
Sex:	female
Recorded:	15 Sep 2026
Case code:	MOCK-042
Earlier measurements:	23 Sep 2025 and 12 Mar 2026

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Software	V4 mockup 0.1
Norms	Mock norms 2026.1 (synthetic)
Recording system	Mitsar amplifier, WinEEG file (mockup label)
Report	R-MOCK-042-0915 v1 · Draft, not issued
Device identifier	UDI: placeholder, assigned at certification
Pipeline	mockup pipeline 0.1.0
Input checksum (sha256)	94184e3a145d4a9294c260f9ff5cc6c611b4e1304fd6f6eedc569dd37249ca08 (synthetic input, not a recording file)
ICA seed	33226
Processed	15 Sep 2026, 10:46, in 400 s

I. Summary

The largest marked deviations from the age norm in this measurement are: relative theta power, eyes open: O1 (z = -4.3), O2 (z = -4.3), Pz (z = -4.0); relative theta power, eyes closed: O1 (z = -4.3), O2 (z = -4.3), Pz (z = -4.1); SMR asymmetry index, eyes open: C3-C4 (z = 4.2); SMR asymmetry index, eyes closed: C3-C4 (z = 4.2); beta1 coherence, eyes closed: T3-C3 (z = 4.1), C3-T5 (z = 4.1), T3-T5 (z = 4.0), T5-O1 (z = 4.0), P3-O1 (z = 4.0), C3-P3 (z = 3.9), T5-P3 (z = 3.8); beta1 coherence, eyes open: Fp2-T4 (z = -3.9), T3-T5 (z = 3.9), T5-O1 (z = 3.9), P3-O1 (z = 3.9), T3-C3 (z = 3.8), C3-P3 (z = 3.8), C3-T5 (z = 3.7), T5-P3 (z = 3.7); P1 latency, Go: O1 (z = 3.7), O2 (z = 3.7). SUM-2

This list ranks the marked values of these analyses by |z|: absolute and relative band power, maps per 1 Hz band, alpha peak frequency, theta/beta ratio, frontal alpha asymmetry index, asymmetry map, coherence per electrode pair, ERP peak amplitude and latency, and VCPT task results. It shows the 5 largest and every marked value of 3 z or more. Not ranked: the spectra per 0.5 Hz bin, which are shown as curves against the norm, and global coherence, which is still to be confirmed. SUM-RULE-2

Marked values in this measurement: band power 6 of 456, maps per 1 Hz band 0 of 1102, alpha peak and ratios 0 of 6, asymmetry map 2 of 96, coherence 15 of 2052, ERP peaks 2 of 21 and VCPT task 0 of 4. SUM-COUNT-1

II. Signal quality and processing

Eyes open: 164 of 180 seconds were analysed after artefact removal. Q-DUR-1

Eyes closed: 171 of 180 seconds were analysed after artefact removal. Q-DUR-1

VCPT: 1168 of 1220 seconds were analysed after artefact removal. Q-DUR-1

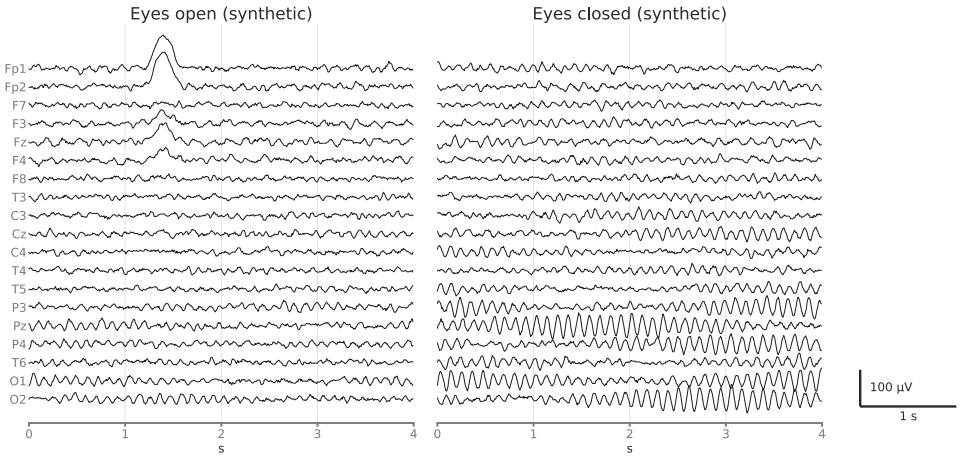
Accepted VCPT trials: Go 95, NoGo 92, Ignore 93, Novelty 95 (of 100 each). Q-TRIALS-1

Artefact removal changed by the clinician: component 6 (muscle, left temporal) was kept although the automatic cleaning proposed removing it; 2 of 19 components were removed. Q-ICA-MAN-1

The analysed segments were confirmed by the clinician as proposed. Q-SEG-AUTO-1

Alpha peak frequency, eyes closed (posterior mean): suggested 9.2 Hz, confirmed unchanged by the clinician; not marked (z = 0.0). MK-AUTO-1

III. Raw EEG excerpt



Synthetic signal, not a recording: the first 4 s of the analysed segments, eyes open and eyes closed, average reference, 19 channels. Scale bars: 100 µV and 1 s.

IV. Anamnesis and clinical interview

Anamnesis	Synthetic example text. Seen for a quantitative EEG and ERP assessment at the family's request; third measurement at this clinic.
History	Synthetic example text. No relevant events reported by the parents.
Medication	Synthetic example text. None reported.
Current symptoms	Synthetic example text. Sleep and appetite as usual, as reported by the parents.

V. Spectra and maps

Eyes open

Figure 1: Absolute power, z against the age norm, Eyes open

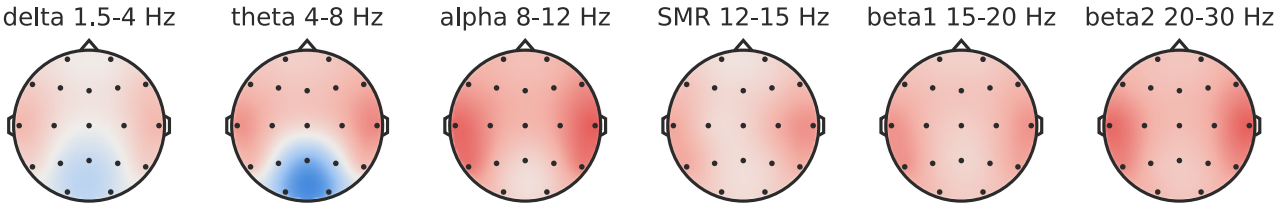
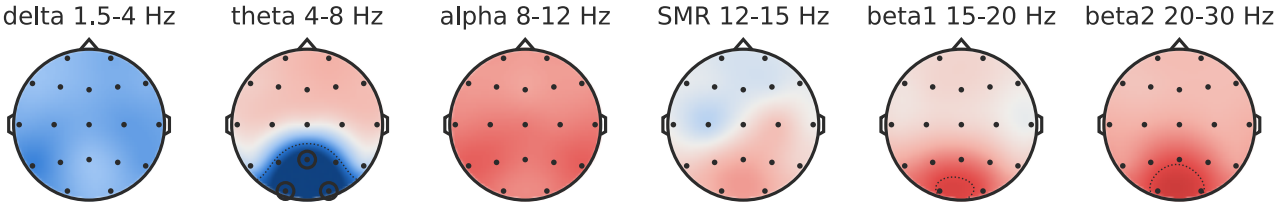


Figure 2: Relative power, z against the age norm, Eyes open



Relative theta power, eyes open, is below the age norm at Pz (z = -4.0), O1 (z = -4.3) and O2 (z = -4.3). SP-REL-1

The alpha peak frequency, eyes open, is 9.2 Hz and is not marked (z = 0.1). APF-IN-2

Eyes closed

Figure 3: Absolute power, z against the age norm, Eyes closed

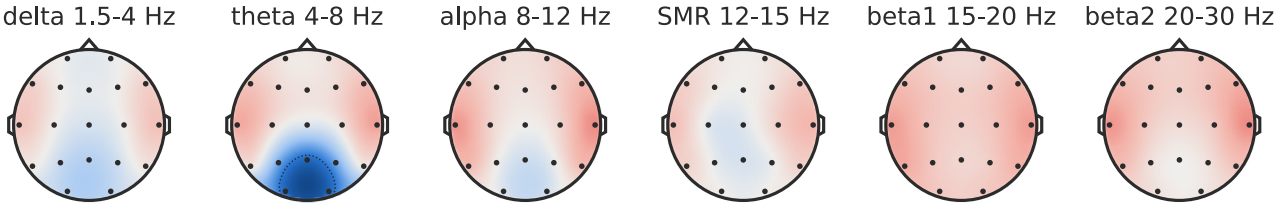
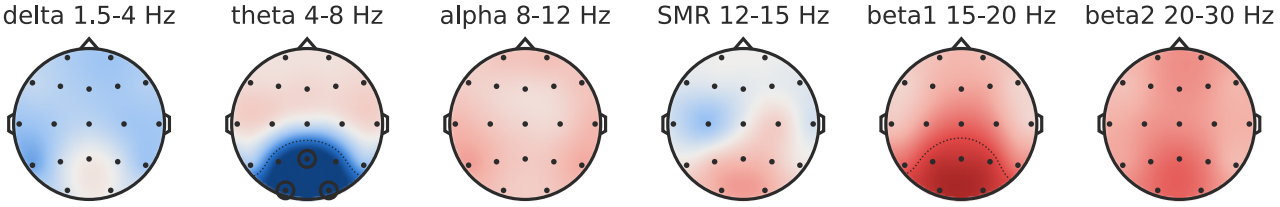
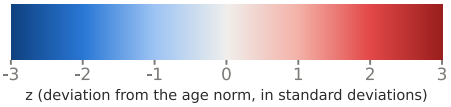


Figure 4: Relative power, z against the age norm, Eyes closed



Relative theta power, eyes closed, is below the age norm at Pz (z = -4.1), O1 (z = -4.3) and O2 (z = -4.3). SP-REL-1

The alpha peak frequency, eyes closed, is 9.2 Hz and is not marked (z = 0.0). APF-IN-2



Ring: marked value (after correction for multiple comparisons).

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

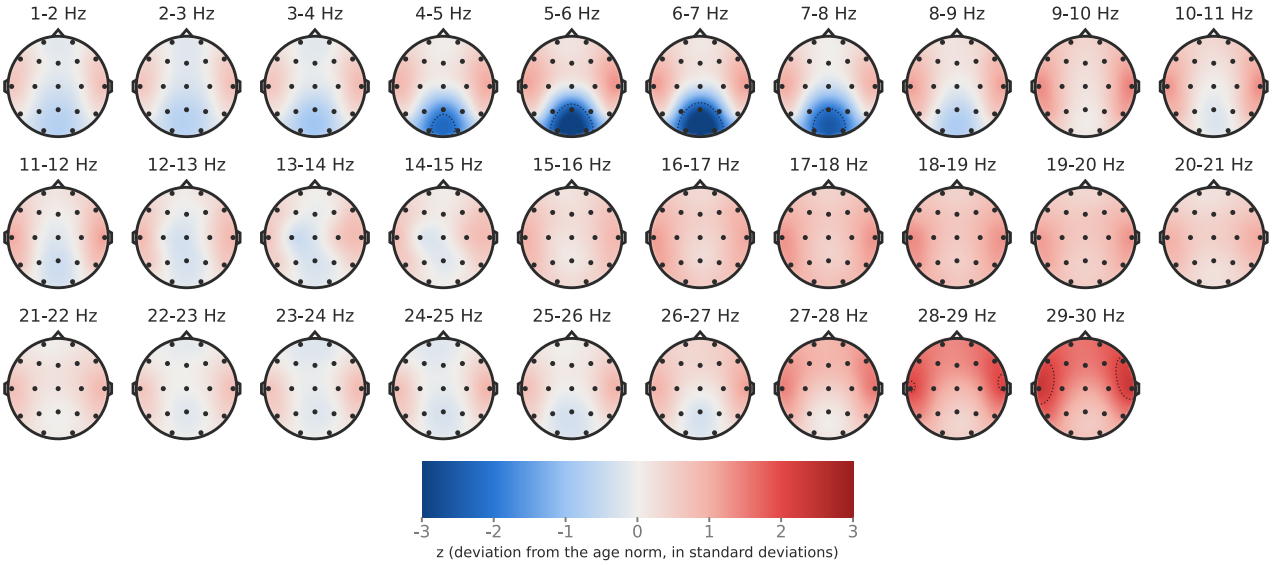
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VI. Maps per 1 Hz band (absolute power, z)

Eyes closed

Figure 5: Maps per 1 Hz band (absolute power, z), Eyes closed



Ring: marked value (after correction for multiple comparisons).

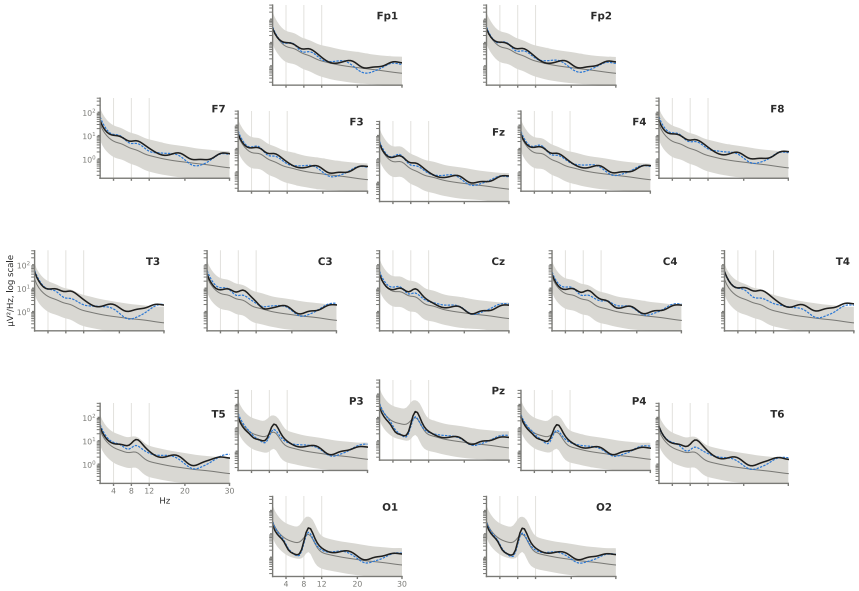
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VII. Absolute spectra per electrode

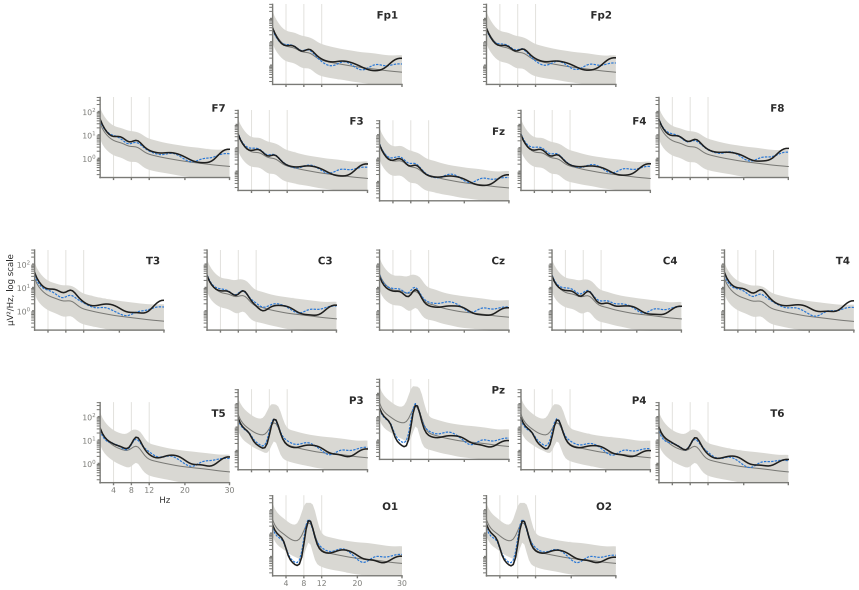
Eyes open

Figure 6: Absolute spectra per electrode, Eyes open



Eyes closed

Figure 7: Absolute spectra per electrode, Eyes closed



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Blue dashed: earlier measurement.

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VIII. Frequency markers

Frequency markers and automatic peak suggestions: core, to confirm

Figure 8: Frequency markers

No.		Electrode	Frequency (Hz)	z	Question, answer	Choice
1	Eyes open	T5	9.0	1.6	-	suggested, confirmed
2	Eyes open	T5	17.5	1.5	-	suggested, confirmed
3	Eyes open	Fz	5.5	0.8	-	suggested, confirmed
1	Eyes closed	T4	9.0	1.5	-	suggested, confirmed
2	Eyes closed	T5	17.0	1.2	-	suggested, confirmed
3	Eyes closed	Fz	5.5	0.3	-	suggested, confirmed

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 9: z at the marker frequency, Eyes open

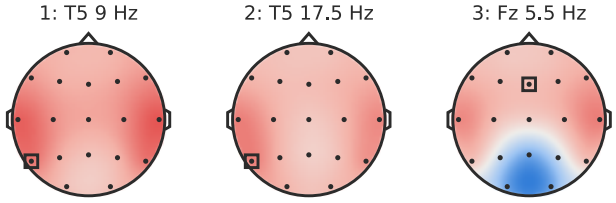
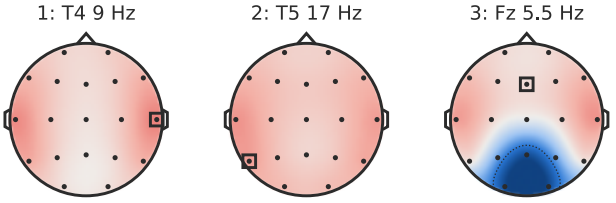


Figure 10: z at the marker frequency, Eyes closed



Square: the electrode of the marker. Ring: marked value (after correction for multiple comparisons).

Frequency markers, eyes open: suggested by the system and confirmed unchanged by the clinician; markers: 3. FM-AUTO-1

Frequency markers, eyes closed: suggested by the system and confirmed unchanged by the clinician; markers: 3. FM-AUTO-1

The system proposes up to 3 peaks of the spectrum per condition, those with the largest positive z (absolute power per 0.5 Hz bin); the clinician confirms, moves, removes or adds markers. Each marker shows its z and whether that value is marked. FM-RULE-1

IX. Band ratios and indices

Measure		Value	Norm mean	z
Theta/beta ratio at Cz	Eyes open	2.74	2.76	0.0
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes open	0.03	-0.01	0.4
Alpha peak frequency (Hz)	Eyes open	9.2	9.1	0.1
Theta/beta ratio at Cz	Eyes closed	1.95	2.82	-1.1
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes closed	0.03	-0.01	0.4
Alpha peak frequency (Hz)	Eyes closed	9.2	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

The theta/beta ratio at Cz, eyes open, is 2.74 (z = 0.0). TBR-1

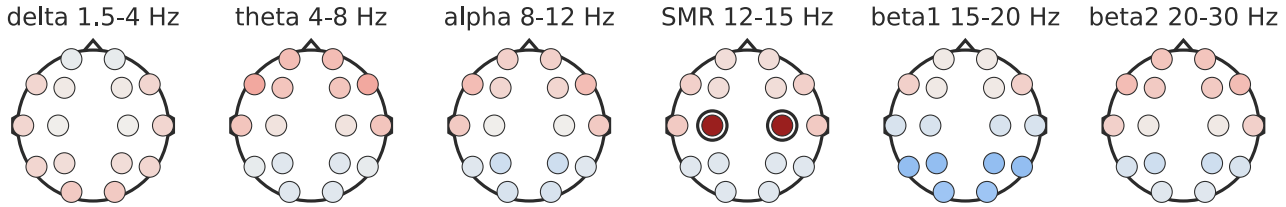
The frontal alpha asymmetry index (ln F4 minus ln F3), eyes open, is 0.03 (z = 0.4). FAA-1

The theta/beta ratio at Cz, eyes closed, is 1.95 (z = -1.1). TBR-1

The frontal alpha asymmetry index (ln F4 minus ln F3), eyes closed, is 0.03 (z = 0.4). FAA-1

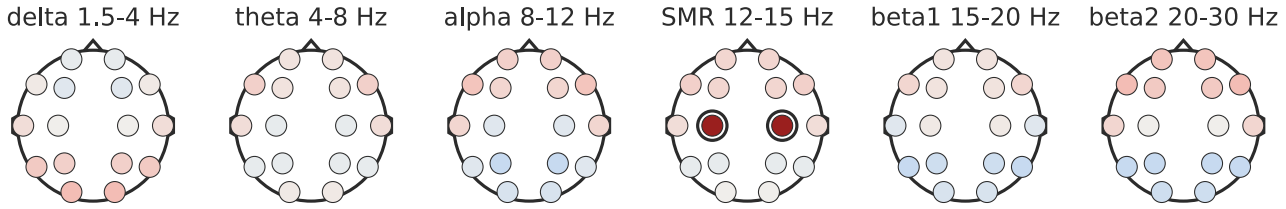
X. Asymmetry

Eyes open

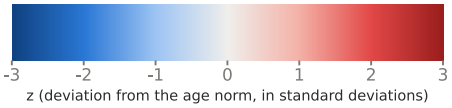


The SMR asymmetry index, eyes open, is above the age norm for C3-C4 (z = 4.2). ASY-1

Eyes closed



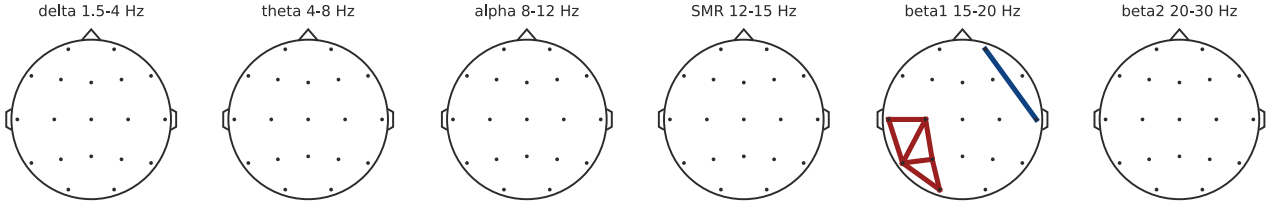
The SMR asymmetry index, eyes closed, is above the age norm for C3-C4 (z = 4.2). ASY-1



Asymmetry index (right minus left) / (right plus left) per electrode pair, z against the age norm. Ring: marked value (after correction for multiple comparisons).

XI. Coherence

Eyes open



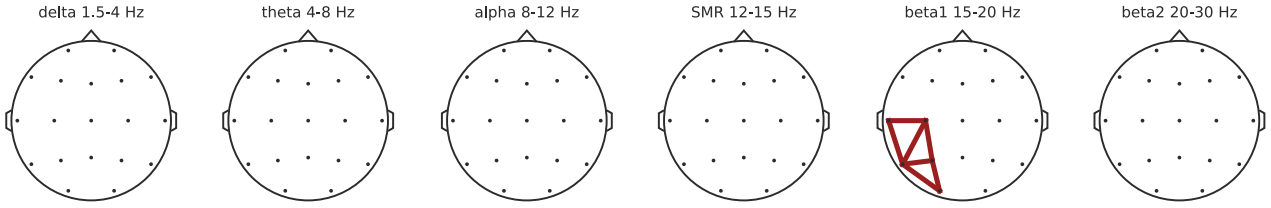
Beta1 coherence, eyes open, is above the age norm for 7 electrode pairs: T3-T5 (z = 3.9), T5-O1 (z = 3.9), P3-O1 (z = 3.9), T3-C3 (z = 3.8), C3-P3 (z = 3.8), C3-T5 (z = 3.7) and 1 more in the table. COH-1

Beta1 coherence, eyes open, is below the age norm for 1 electrode pair: Fp2-T4 (z = -3.9). COH-1

Figure 11: Marked electrode pairs, Eyes open

Band	beta1							
Pair	Fp2-T4	T3-T5	T5-O1	P3-O1	T3-C3	C3-P3	C3-T5	T5-P3
Coherence	0.01	0.62	0.62	0.71	0.67	0.64	0.59	0.70
z	**-3.9**	**3.9**	**3.9**	**3.9**	**3.8**	**3.8**	**3.7**	**3.7**

Eyes closed



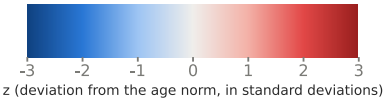
Beta1 coherence, eyes closed, is above the age norm for 7 electrode pairs: T3-C3 (z = 4.1), C3-T5 (z = 4.1), T3-T5 (z = 4.0), T5-O1 (z = 4.0), P3-O1 (z = 4.0), C3-P3 (z = 3.9) and 1 more in the table. COH-1

Figure 12: Marked electrode pairs, Eyes closed

Band	beta1						
Pair	T3-C3	C3-T5	T3-T5	T5-O1	P3-O1	C3-P3	T5-P3
Coherence	0.68	0.62	0.63	0.63	0.71	0.66	0.71
z	**4.1**	**4.1**	**4.0**	**4.0**	**4.0**	**3.9**	**3.8**

Global coherence (mean over all pairs), to confirm

	delta	theta	alpha	SMR	beta1	beta2
Eyes open, Value	0.16	0.18	0.23	0.22	0.21	0.17
Eyes open, z	-2.2	-1.9	-0.3	0.0	0.1	-0.6
Eyes closed, Value	0.18	0.19	0.24	0.22	0.21	0.19
Eyes closed, z	-1.6	-1.5	-0.7	-0.1	0.1	-0.1



Lines: marked electrode pairs, coloured by z. Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z. Global coherence is a core function still to be confirmed; it is not ranked in the summary.

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

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XII. Event-related potentials at the electrodes (channel ERPs)

Figure 13: Event-related potentials at the electrodes (channel ERPs): Go

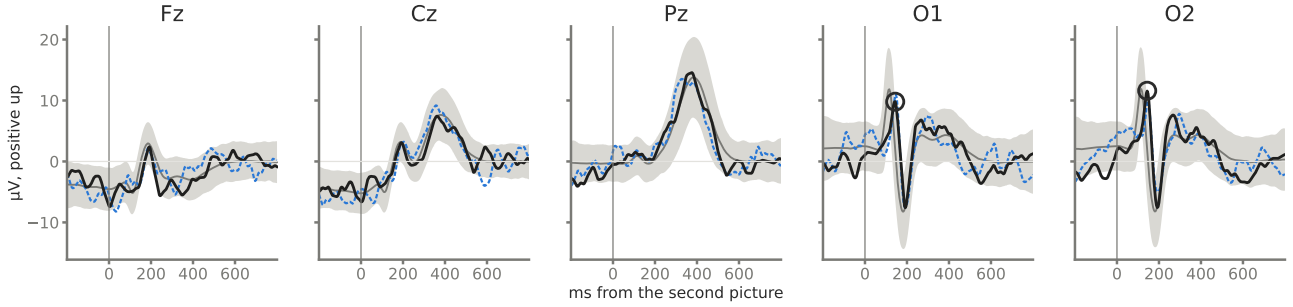
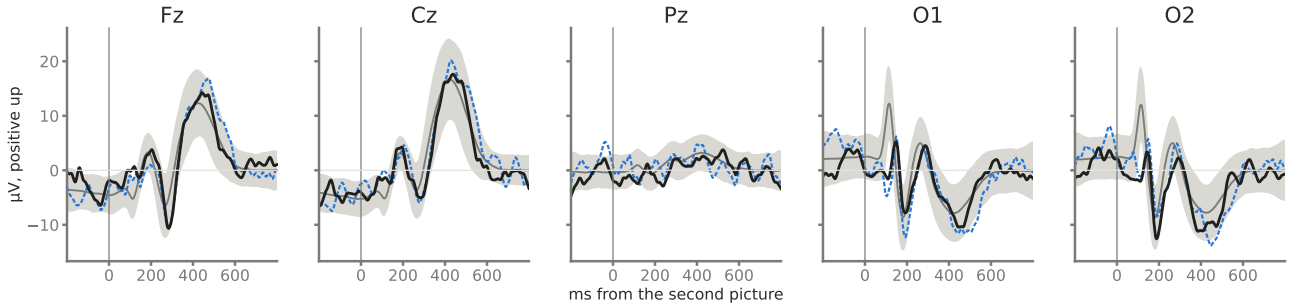


Figure 14: Event-related potentials at the electrodes (channel ERPs): NoGo



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Time 0: second stimulus. Ring: marked value (after correction for multiple comparisons). Blue dashed: earlier measurement.

Figure 15: ERP peak values

Component	Trial type	Electrode	Amplitude (µV)	z	Latency (ms)	z
P1	Go	O1	9.8	-1.0	144	**3.7**
P1	Go	O2	11.6	-0.5	144	**3.7**
N1	Go	O1	-7.6	0.6	192	0.9
N1	Go	O2	-7.6	0.6	192	1.0
N2	NoGo	Fz	-10.7	-1.1	284	1.2
N2	NoGo	Cz	-5.0	-0.2	280	1.1
P3	Go	Pz	14.6	-0.1	372	-0.4
P3	NoGo	Cz	17.6	-0.2	440	0.5
P3	NoGo	Fz	14.3	0.2	444	0.6
P3a	Novelty	Cz	16.4	0.4	360	1.2
CNV	Go	Cz	-4.8	0.0	-200 to 0	

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

ERP peaks, Go trials: the suggested peaks were confirmed unchanged by the clinician. ERP-PEAK-AUTO-1

ERP peaks, NoGo trials: the suggested peaks were confirmed unchanged by the clinician. ERP-PEAK-AUTO-1

ERP peaks, Novelty trials: the suggested peaks were confirmed unchanged by the clinician. ERP-PEAK-AUTO-1

P1 latency at O1, Go trials, is later than the age norm (144 ms, z = 3.7). ERP-LAT-1

P1 latency at O2, Go trials, is later than the age norm (144 ms, z = 3.7). ERP-LAT-1

XIII. Checklist over channel ERPs

Checklist over channel ERPs: core, to confirm (proposal for the clinical lead)

Component	Trial type	Electrode	Window (ms)	Latency earlier	Latency later	Amplitude larger	Amplitude smaller	Shape differs
P1	Go	O1	70 to 170		✓			
P1	Go	O2	70 to 170		✓			
N1	Go	O1	140 to 250					
N1	Go	O2	140 to 250					
N2	NoGo	Fz	210 to 330					
N2	NoGo	Cz	210 to 330					
P3	Go	Pz	280 to 560					
P3	NoGo	Cz	300 to 600					
P3	NoGo	Fz	300 to 600					
P3a	Novelty	Cz	250 to 420					
CNV	Go	Cz	-200 to 0	-	-			

Ticks by the clinician, against the norm curves of the channel ERP figures. For N1, N2 and CNV, larger means more negative. A dash: item not offered for this row.

Checklist over channel ERPs, P1 at O1, Go trials (70 to 170 ms), ticked by the clinician: later latency than the norm curve. ERPCL-TICK-1

Checklist over channel ERPs, P1 at O2, Go trials (70 to 170 ms), ticked by the clinician: later latency than the norm curve. ERPCL-TICK-1

Checklist over channel ERPs: rows without a tick: 9. ERPCL-REST-1

In this checklist the clinician ticks what the channel ERP figures show against the norm curves; the system computes nothing from the ticks and states them as ticked. ERPCL-RULE-1

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

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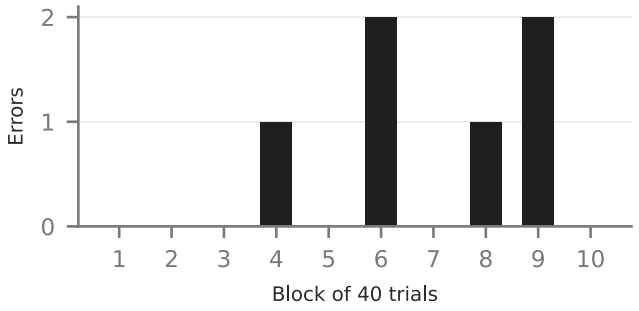
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XIV. VCPT task performance

Measure	Value	Percentile	z
Omission errors	2	46	-0.1
Commission errors	4	81	0.9
Mean reaction time	409 ms	47	-0.1
Reaction time variability (SD)	86 ms	42	-0.2

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 16: Errors per block of 40 trials (no norm)



Omission errors: 2, not marked (percentile 46). BEH-IN-2
Commission errors: 4, not marked (percentile 81). BEH-IN-2
Mean reaction time: 409 ms, not marked (percentile 47). BEH-IN-2
Reaction time variability (SD): 86 ms, not marked (percentile 42). BEH-IN-2

XV. Comparison of measurements

Measure	Earlier 23 Sep 2025 (uploaded later)	z	Earlier 12 Mar 2026	z	Current 15 Sep 2026	z
Omission errors	3	0.3	3	0.3	2	-0.1
Commission errors	6	1.5	5	1.2	4	0.9
Mean reaction time	431 ms	0.2	421 ms	0.1	409 ms	-0.1
Reaction time variability (SD)	97 ms	0.2	95 ms	0.2	86 ms	-0.2
Theta/beta ratio at Cz, Eyes open	2.22	-0.7	2.61	-0.2	2.74	0.0
Alpha peak frequency (Hz), Eyes open	8.9	-0.2	9.1	0.0	9.2	0.1
Theta/beta ratio at Cz, Eyes closed	2.09	-0.9	1.93	-1.1	1.95	-1.1
Alpha peak frequency (Hz), Eyes closed	9.1	-0.1	9.1	-0.1	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 17: Earlier 23 Sep 2025: Absolute power, z against the age norm, Eyes closed

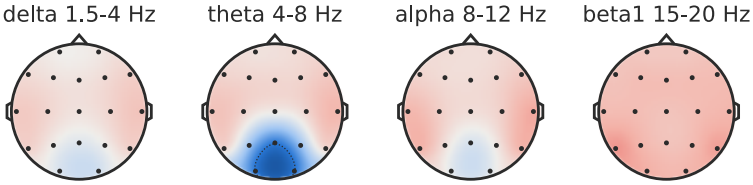


Figure 18: Earlier 12 Mar 2026: Absolute power, z against the age norm, Eyes closed

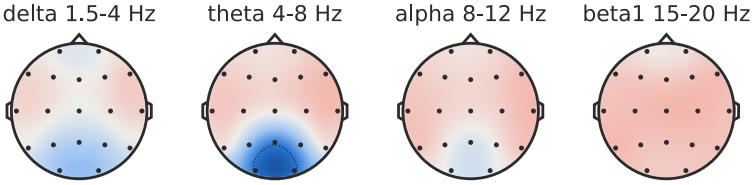
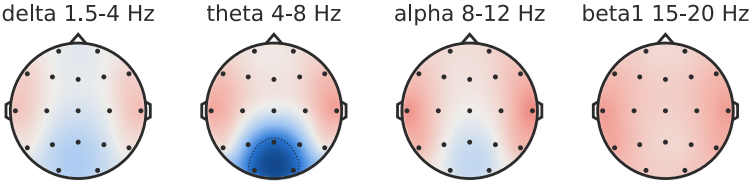


Figure 19: Current 15 Sep 2026: Absolute power, z against the age norm, Eyes closed



Ring: marked value (after correction for multiple comparisons).

Beta1 coherence T3-C3, eyes closed: z = 4.7 on 23 Sep 2025, z = 4.1 on 12 Mar 2026 and z = 4.1 on 15 Sep 2026. CMP-N-1

Beta1 coherence T3-T5, eyes closed: z = 4.6 on 23 Sep 2025, z = 4.3 on 12 Mar 2026 and z = 4.0 on 15 Sep 2026. CMP-N-1

Beta1 coherence T5-O1, eyes closed: z = 4.6 on 23 Sep 2025, z = 4.0 on 12 Mar 2026 and z = 4.0 on 15 Sep 2026. CMP-N-1

Beta1 coherence P3-O1, eyes closed: z = 4.6 on 23 Sep 2025, z = 4.2 on 12 Mar 2026 and z = 4.0 on 15 Sep 2026. CMP-N-1

Beta1 coherence C3-P3, eyes closed: z = 4.5 on 23 Sep 2025, z = 4.2 on 12 Mar 2026 and z = 3.9 on 15 Sep 2026. CMP-N-1

Beta1 coherence T5-P3, eyes closed: z = 4.5 on 23 Sep 2025, z = 4.3 on 12 Mar 2026 and z = 3.8 on 15 Sep 2026. CMP-N-1

The earlier measurement of 23 Sep 2025 was processed again with the current software and norms (V4 mockup 0.1; Mock norms 2026.1 (synthetic)). L-REPROC-1

The earlier measurement of 12 Mar 2026 was processed again with the current software and norms (V4 mockup 0.1; Mock norms 2026.1 (synthetic)). L-REPROC-1

Measurements are shown side by side, each against its own age norm; differences between them are not tested for statistical significance. L-SIDE-1

Measurements uploaded later are processed from their recordings; reports and numbers made by earlier software are not imported. L-IMPORT-1

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XVI. Notes on the results

The results are one input to the clinician's assessment. The software does not diagnose and does not recommend treatment. L-AID-1

z is the distance of a value from the normative mean, in standard deviations of the normative sample; about 95% of the normative sample lies between z = -2 and z = 2. L-Z-1

Marked values differ from the age norm after correction for multiple comparisons, which keeps chance results among the many comparisons rare; a value beyond 2 z that is not marked may deviate by chance alone. The method of correction is still to be chosen. L-MC-1

XVII. Method and norms

Recording: Mitsar amplifier, WinEEG file (mockup label); 19 scalp electrodes of the 10-20 system (Fp1 to O2); conditions: eyes open, eyes closed and VCPT. MET-REC-1

Processing: average reference of the 19 scalp channels; mains notch filter at 50 Hz; analysis rate 250 Hz; artefacts removed by independent component analysis (ICA) with reference maps, then remaining artefact segments excluded. MET-PROC-1

Spectra: 1 to 30 Hz in steps of 0.5 Hz, compared with the norm as log10 power. Bands: delta 1.5 to 4 Hz, theta 4 to 8 Hz, alpha 8 to 12 Hz, SMR 12 to 15 Hz, beta1 15 to 20 Hz and beta2 20 to 30 Hz. Theta/beta ratio at Cz with theta 4 to 8 Hz and beta 13 to 21 Hz. MET-SPEC-1

ERPs: VCPT with 100 trials each of Go, NoGo, Ignore and Novelty; epochs from -200 to 800 ms around the second picture, baseline before the cue; peaks read off the curves in fixed windows. MET-ERP-1

Norms: Mock norms, version 2026.1 (synthetic); age range of the norms 6 to 67 years; compared at the age at recording, 10 years. MET-NORM-1

Marking: a value is marked when it differs from the age norm after correction for multiple comparisons; until the method is chosen this mockup uses the Benjamini-Hochberg false discovery rate (q = 0.05) within each analysis block. MET-MC-1

Recording conditions (documented at upload)

Time of recording	09:40
Medication on the recording day	none reported
Sleep the night before	as usual
Drowsiness observed during the recording	no

XVIII. Interpretation

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

XIX. Conclusion

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

Issued by

Issued by	_____ (entered when the clinician issues the report)
Date of issue	_____ (entered when the clinician issues the report)
Version	1 · Draft, not issued
Replaces	none, first version

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